

433 HW1

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Problem 1

```
seq(0,1.6,.4) # used sequence command with (starting value, end value, increment)

## [1] 0.0 0.4 0.8 1.2 1.6
```

Problem 2

```
0:80/20 # create sequence from 0 to 80, divide each value by 20 so that

## [1] 0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50 0.55 0.60 0.65 0.70
## [16] 0.75 0.80 0.85 0.90 0.95 1.00 1.05 1.10 1.15 1.20 1.25 1.30 1.35 1.40 1.45
## [31] 1.50 1.55 1.60 1.65 1.70 1.75 1.80 1.85 1.90 1.95 2.00 2.05 2.10 2.15 2.20
## [46] 2.25 2.30 2.35 2.40 2.45 2.50 2.55 2.60 2.65 2.70 2.75 2.80 2.85 2.90 2.95
## [61] 3.00 3.05 3.10 3.15 3.20 3.25 3.30 3.35 3.40 3.45 3.50 3.55 3.60 3.65 3.70
## [76] 3.75 3.80 3.85 3.90 3.95 4.00
```

each number in the sequence is separated by increments of 0.05.

Problem 3

```
n = 100 # define n as 100
n*(n+1)/2 # use formula for summing integers, output is the sum of the first 100 integers

## [1] 5050
```

Problem 4

```
length(seq(6, 55, 4/7)) # create sequence, then determine how many values are in

## [1] 86
```

```
# the sequence using length function
```

Problem 5

```
x = seq(1,100) # create sequence from 1 to 100  
sum(1/x^2) # apply transformations to the sequence, squaring each value then
```

```
## [1] 1.634984
```

```
# taking the reciprocal, use sum function to add each value together  
# result is close to pi^2/6, which is approximately 1.6449
```

Problem 6

```
A = matrix(c(1, 6, 3, 2, 3, 1, 5, 2, 1), 3, 3, byrow = TRUE) # create A matrix  
B = matrix(c(3, 5, 4, 2, 1, 7, 1, 0, 7), 3, 3, byrow = TRUE) # create B matrix  
A%*%B # Part a, multiply matrices together
```

```
##      [,1] [,2] [,3]  
## [1,]   18   11   67  
## [2,]   13   13   36  
## [3,]   20   27   41
```

```
B%*%A
```

```
##      [,1] [,2] [,3]  
## [1,]   33   41   18  
## [2,]   39   29   14  
## [3,]   36   20   10
```

```
solve(A)%*%B%*%t(A) # Part b, use solve and t function to find inverse and transpose
```

```
##      [,1]      [,2] [,3]  
## [1,]   1.5  0.1428571  0.5  
## [2,]  11.5  5.4285714  8.5  
## [3,]  -8.5 -2.5714286 -7.5
```

```
# of matrix A, respectively, then perform matrix multiplication  
A%*%A%*%A%*%A # Part c, use repeated multiplication to acquire fourth power of matrix A
```

```
##      [,1] [,2] [,3]  
## [1,] 1342 1986  852  
## [2,]  803 1299  566  
## [3,] 1138 1978  872
```

```

b = matrix(c(3, 4, 2), 3, 1, byrow = FALSE) # Part d, create the matrix for b
# Bx = b
# (B^-1)*Bx=(B^-1)*b, multiply both sides by the inverse matrix of B
# Ix = (B^-1)*b, the inverse of a matrix multiplied by its original is the identity matrix
# x = (B^-1)*b, Identity of x is simply x.
solve(B)%*%b # this output is the solution for x, the values which solve the equation

```

```

##           [,1]
## [1,]  3.166667
## [2,] -1.166667
## [3,] -0.166667

```